

Gustav Robert Kirchhoff's Voltage Law



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A loop is any path through a circuit that starts and ends at the same point. Consider, for example, the Ampeg B-22-X high-gain input circuit. Here are three possible loops:

guitar Loop 2 Loop 1 Loop 3

These are not all the loops in the circuit. For example the path from ground through the guitar, through the capacitor, through the right-most resistor, and back to ground is also a loop.

Gustav Robert Kirchhoff's Voltage Law

According to Kirchhoff's Voltage Law, or KVL as it is known, the sum of the voltage rises around a closed loop equals the sum of the voltage drops.



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Problem

Show that Kirchhoff's Voltage Law applies to one of the ripple filters in a Traynor YBA-3.

Loop 1 420V 375V

Solution

There is only one closed loop shown. Starting at the 420V screen supply and working clockwise, there is a voltage drop of 45V to 375V. Then there is a voltage drop of 375V to 0V. Finally there is a voltage rise of 420V to 420V. Thus there is only one voltage rise and it equals the sum of the voltage drops:

$$420V = 45V + 375V$$

For any other starting point, clockwise or counterclockwise, KVL also holds.

KVL in Practice

Kirchhoff's Voltage Law is used often in circuit analysis because it allows us to determine an unknown voltage based on known voltages. The choice of which loop to use is a matter of convenience.

Problem

Given that the cathode-to-ground voltage V_K for an Ampeg B-42-X preamp tube is +2.5V, what is the grid-to-cathode voltage V_{GK} ? The grid-to-ground voltage V_G is 0V.

Loop 1 RG RK G K

Solution

We can form a counterclockwise loop from the grid, through the grid resistor R_G to ground, through the cathode resistor R_K to the cathode, and finally from the cathode to the grid. In this loop there are two voltage rises, the known voltage $V_K = 2.5V$ and the unknown voltage V_{GK} from cathode to grid. There are no voltage drops. According to Kirchhoff's Voltage Law

$$V_{GK} + 2.5V = 0V$$

This means

$$V_{GK} = -2.5V$$

The voltage of the grid relative to the cathode, is thus -2.5V.

This is a very simple loop and we could easily have determined the answer just by looking at the circuit: if the grid is at 0V relative to ground and the cathode is at +2.5V relative to ground then the grid-to-cathode voltage must be -2.5V. Try using simple logic like this with a Fender Bassman 5F6-A or Hiwatt DR103 tone stack! KVL gives us a formal, step-by-step procedure to keep from making mistakes, no matter how complicated the circuit gets.

Problem

Your Princeton 5F2 clone uses an output transformer that is, like all transformers, far from perfect. For DC voltages its resistance should ideally be zero, but using your vintage Hewlett Packard Model 410B vacuum tube voltmeter you measure an internal winding resistance of 50Ω . The plate supply voltage is 305V and the voltage across the 470Ω cathode resistor is 19V. What is the plate current that flows from the 305V supply, through the transformer, through the tube, and finally through the cathode resistor? What is the 6V6GT plate voltage, measured from plate to cathode? (For this problem, we assume that screen current is negligible.)

305V 19V 470Ω + _

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